IMPERIAL

MSc Individual Project

Imperial College London

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A Vision-Guided Pre-Palpation Support Pipeline for Phantom Surface Exploration in Simulation

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September 11, 2025

Abstract

Tactile perception plays a crucial role in robotic-assisted surgery, as subtle changes in tissue stiffness or surface geometry can reveal tumours that may be missed by conventional imaging. However, most existing approaches assume that a probe is already correctly placed, leaving open the critical challenge of how to safely and efficiently prepare for palpation.

This project aims to develop a vision-guided *pre-palpation support* pipeline that ensures phantoms are accurately localised, their boundaries reconstructed, and safe regions identified before any probing begins.

The proposed pipeline integrates five stages—phantom perception, boundary construction with closure verification, surface scanning, Digital Elevation Model (DEM) reconstruction, and anomaly detection leading to palpation suggestions—implemented and tested in the Genesis physics simulator.

Experimental results showed that depth-based clustering provided more robust perception than colour-based segmentation, and RGB-D gradient fusion produced denser and more accurate boundary reconstructions than Canny edges. Quantitative evaluation using Chamfer and Hausdorff distances confirmed boundary accuracy, while additional stability measures validated consistent performance across varied conditions.

In conclusion, the system bridges vision-based perception with tactile exploration by providing both a safe scanning region and actionable palpation waypoints. This pre-palpation support framework improves the efficiency, safety, and reliability of robotic-assisted tumour detection, and establishes a foundation for future integration with more realistic tissue models and physical tactile sensors.

Acknowledgements

I would like to express my deepest gratitude to my supervisor, **Dr. Dandan Zhang**, for her invaluable guidance, encouragement, and support throughout this project. Her insights and constructive feedback have been instrumental in shaping both the direction and quality of my work.

I am also sincerely grateful to **Fan Wen**, whose generous help and advice during the project provided me with technical guidance and inspiration.

My heartfelt thanks go to my parents, whose unwavering support and understanding have sustained me throughout my studies. I am equally grateful to all the teachers who have taught and supported me during my time at Imperial College, contributing to my academic growth.

I would also like to thank myself for the persistence and effort dedicated to this project. Finally, I wish to acknowledge the **Genesis development team** for making their simulator open-source, which provided a powerful foundation for this work.

I also wish to thank the rapid advancement of technology, which made tools and resources accessible for completing this research, and my friend **Cha**, whose companionship and encouragement have accompanied me throughout this journey.

Beyond academia, I am grateful to **Ultimate Frisbee**, which has helped me recover energy, clear my mind, and find fresh motivation to continue debugging and developing throughout this project.

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Chapter 1

Introduction

1.1 Background

Tactile perception plays a critical role in robotic-assisted diagnosis and surgery, particularly in detecting abnormalities such as tumours. While vision-based modalities such as MRI and ultrasound imaging are widely employed, haptic sensing provides complementary information that is especially valuable for identifying early-stage tumours. Subtle variations in stiffness, elasticity, or texture may not be readily discernible in conventional images but can be revealed through palpation. Despite this potential, collecting real-world tactile datasets for algorithm development remains ethically, logistically, and financially challenging.

Recent advances in physics-based simulation provide an opportunity to overcome these challenges. Among them, Genesis offers high-fidelity soft-body interactions, differentiable physics, and generative data creation capabilities. These characteristics make it an ideal platform for replicating tissue–probe interactions and producing synthetic tactile datasets for machine learning.

1.2 Motivation

The motivation for this study is to provide a reliable *pre-palpation support* system rather than to replace palpation itself. In clinical and robotic contexts, palpation remains the gold standard for confirming tissue abnormalities through direct force feedback. However, palpation is inefficient and potentially unsafe if the probe is not properly positioned or if the scanning area is incomplete. Our approach therefore focuses on creating a vision-guided pipeline that establishes phantom perception, boundary construction, and surface coverage before contact. The output of this pipeline is not an autonomous decision but **palpation suggestions**—concrete candidate regions that should be probed. In this way, the system serves as an assistive tool that improves the efficiency, safety, and reliability of robotic-assisted tumour detection while leaving the final decision to tactile confirmation.

Ultimately, the goal of this project is to integrate complementary computer vision methods within a simulation framework so that palpation can be initiated from a stable, well-defined starting point, ensuring that subsequent tactile exploration is both targeted and clinically meaningful.

1.3 Related work

Early robotic systems such as the *da Vinci Surgical System* enhanced dexterity and vision but lacked integrated haptic sensing, limiting intraoperative tumour detection through palpation [11, 16]. To address this limitation,

researchers introduced tactile sensors ranging from piezoelectric devices to multimodal designs capable of measuring force distribution, viscoelastic properties, and surface texture [5, 30, 4, 15, 3]. Vision-based tactile sensors (VBTSs) emerged as a particularly powerful direction: they translate deformation into images via internal cameras and elastomeric skins, allowing high-resolution reconstruction of contact geometry and force distribution [31]. Representative designs include TacTip, which uses a dome-shaped elastomer with internal markers to capture contact events [13]; OmniTact, which extends vision-based sensing to curved surfaces using multiple micro-cameras [17]; and ViTacTip, which integrates visual and tactile modalities into a single probe, providing robust performance under variable lighting [8].

In parallel, physics-based simulation platforms have been widely adopted to generate synthetic tactile datasets, mitigating the ethical, logistical, and cost barriers of real-world data collection [6]. SOFA provides accurate finite-element soft-tissue modelling but at high computational cost [29]; MuJoCo offers fast rigid-body dynamics, though it is less specialised for deformable tissues [25]; and Genesis combines high-fidelity soft-body physics with differentiable simulation, making it particularly suited for AI-driven tactile perception [1]. These platforms have facilitated research in contact modelling, synthetic data generation, and tumour classification from tactile signals.

Despite these advances, most existing studies evaluate modules in isolation. Many tumour detection algorithms assume that the probe is already placed correctly on the tissue [22, 9, 27, 6], overlooking the critical precondition of localising the phantom, verifying boundary closure, and ensuring surface coverage before palpation. Without this step, palpation risks becoming inefficient or unsafe, as the probe may miss regions of interest or apply contact outside the target. Similarly, vision-based approaches—such as colour segmentation, depth clustering [21], and edge detection—highlight trade-offs between efficiency and robustness, but are rarely embedded into an end-to-end workflow that explicitly serves palpation preparation. Consequently, the transformation of vision outputs into actionable palpation suggestions remains underexplored.

1.4 Aim & Objectives

Aim The primary aim of this project is to ensure that, before palpation begins, the probe can be reliably guided to the correct location above the target and that the surface is safely and completely covered. Rather than replacing palpation, the system serves as a pre-palpation support pipeline: it first localises the phantom, reconstructs its boundary, verifies surface coverage, and then generates palpation suggestions—specific points or regions where the probe should touch. In this way, tactile exploration can start from a stable and meaningful position, making subsequent palpation safer and more efficient.

Objectives To achieve the overall aim, this project is structured around the following objectives, each corresponding to a distinct stage of the proposed pipeline:

- **Perception (Phantom localisation):** Implement and evaluate two complementary vision-based localisation methods—HSV+Depth segmentation and depth-based clustering (DBSCAN)—to estimate the centroid and spatial extent of the target under varying illumination conditions. This provides the probe with a reliable initial position for subsequent stages.
- **Boundary construction:** Employ Canny edge detection and RGB-D gradient fusion to reconstruct the outer contour of the phantom, thereby defining a safe region for interior surface scanning.
- **Closure verification:** Develop and apply a hybrid verification scheme combining image-plane flood-fill

with graph-based cycle detection to confirm whether the accumulated boundary forms a valid closed loop, while disambiguating nested contours caused by local surface protrusions.

- **Surface scanning and DEM reconstruction:** Perform dense surface sampling within the verified safe region and interpolate the resulting point cloud into a digital elevation model (DEM) [10] to ensure uniform coverage of the phantom's surface.
- **Anomaly detection and palpation suggestions:** Detect protrusion-like anomalies from the DEM, characterise them by centroid, elevation, and area, and translate the most prominent candidates into prioritised palpation waypoints.
- **Evaluation framework:** Establish a unified evaluation protocol to assess the entire pipeline in terms of localisation error, boundary closure quality, surface coverage completeness, success rate, and efficiency, enabling systematic comparison between methods.

1.5 Contributions

This project contributes a vision-guided pre-palpation framework designed to ensure that a tactile sensor mounted on a robotic arm can be guided reliably to the phantom and provided with meaningful palpation locations. A complete five-stage pipeline was implemented and validated (Fig. 2.1), integrating perception, motion control, boundary construction, surface scanning, and anomaly detection into a coherent workflow that culminates in palpation suggestions. Within this framework, two localisation strategies—HSV+Depth segmentation and depth-based clustering—were compared, with depth clustering ultimately selected for its superior robustness under variable illumination. For boundary construction, Canny edge detection was assessed against RGB-D gradient fusion, with the latter chosen for its ability to generate denser and more continuous contours, thereby defining safer regions for subsequent scanning. A hybrid closure verification strategy was introduced to eliminate incomplete or nested loops and to ensure that only the outermost valid boundary was accepted. Within this validated region, surface points were densely sampled and interpolated into DEMs, from which local protrusions were detected and converted into prioritised palpation waypoints. Together, these contributions establish a coherent pre-palpation pipeline that not only reconstructs the phantom surface with high fidelity but also translates geometric anomalies into concrete palpation suggestions, thereby ensuring that tactile exploration is safe, accurate, and efficient.

Chapter 2

Method

2.1 Pipeline Overview

Reliable palpation for tumour detection requires the probe to be positioned accurately and to maintain stable coverage of the tissue surface. We therefore implement a five-stage vision-based servo loop—**Perception** → **Pose Initialisation & Motion Control** → **Boundary Construction** → **Surface Scanning** → **Anomaly Detection & Palpation Suggestions**—that first localises the phantom, then reconstructs a closed boundary to delimit a safe scanning region, interpolates the surface into a DEM, and finally extracts tumour-like regions of interest as palpation targets (Fig. 2.1).

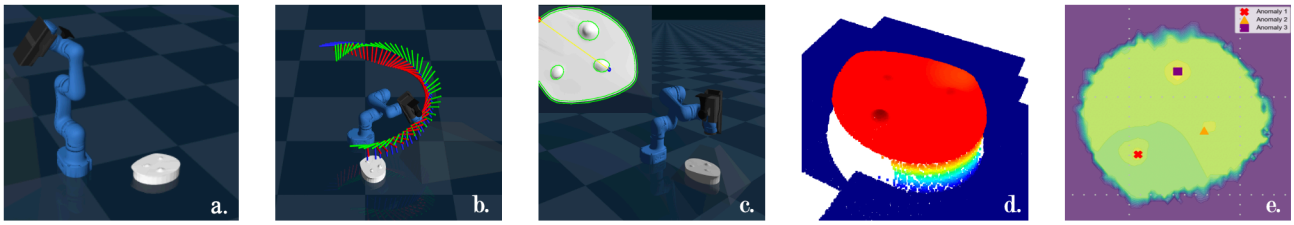

Figure 2.1: **Pipeline overview for vision-guided tactile perception.** (a) **Perception**: 360° RGB-D sweep and phomtom localisation. (b) **Pose Initialisation & Motion Control**: IK-based positioning above the phantom. (c) **Boundary Construction**: edge-based contour extraction and closure verification. (d) **Surface Scanning & DEM Reconstruction**: interior point-cloud sampling and height-map generation. (e) **Anomaly Detection & Palpation Suggestions**: protrusion detection and ROI selection for palpation.

In the **Perception** stage, the manipulator executes a 360° sweep and acquires RGB-D frames from predefined viewpoints. Two complementary localisation methods are evaluated: HSV segmentation with depth back-projection, and depth-based clustering (DBSCAN). Both provide estimates of the phantom centroid and lateral extent, which seed the subsequent stages.

In **Pose Initialisation & Motion Control**, Genesis' inverse kinematics (IK) with trajectory interpolation guides the eye-in-hand camera to a pre-scan height above the estimated centroid, ensuring smooth, collision-free motion and sufficient positional accuracy for exploration.

In **Boundary Construction**, contour points are extracted from RGB-D frames using either (i) a Canny edge detector or (ii) an RGB-D fusion of intensity and depth gradients. These points are back-projected into 3D and accumulated as the arm advances along the detected contour. Closure verification applies a hybrid criterion to the accumulated boundary cloud: an image-plane flood-fill area test and a graph-based k-nearest-neighbour cycle test. When multiple closed loops appear (e.g., due to protrusions), the *outermost* loop (maximum projected area)

is retained, and closure is accepted only if its area is consistent with the Stage-1 extent prior. The resulting closed polygon is designated as the *safe region* for interior scanning.

In **Surface Scanning**, the safe region is rasterised into a uniform grid of sampling locations. At each position the manipulator hovers to capture RGB-D frames, which are fused into a point cloud. To regularise density, the point cloud is interpolated onto a grid, producing a DEM of surface height suitable for neighbourhood-based analysis.

Finally, in **Anomaly Detection & Palpation Suggestions**, local protrusions on the DEM are identified using adaptive thresholding and refined through connected-component analysis. Each anomaly is characterised by centroid, elevation, and area, and the most prominent candidates are converted into probing waypoints (approach poses, hover heights, indentation limits). These regions of interest form the actionable palpation targets for subsequent tactile exploration.

The full source code for all modules of the pipeline is provided in Appendix A.2.

2.2 Simulation Environment

All experiments were conducted in the Genesis physics simulator [1], which provides GPU-accelerated dynamics, differentiable physics, and integrated robotic control interfaces. Simulation was chosen at this stage for three key reasons. First, it offers a safe and controllable environment for developing vision-based servoing without the risks of hardware damage or probe–tissue collisions. Second, it allows systematic variation of lighting, phantom geometry, and placement, which is essential for evaluating robustness and generalisability. Finally, the same URDF robot models and camera configurations can be reused in physical experiments, ensuring a clear path towards sim-to-real transfer. Thus, simulation serves not only as a prototyping tool but also as a foundation for validating palpation-oriented pipelines under reproducible conditions.

Scene configuration. The world frame was defined with standard gravity $(0, 0, -9.81\,\mathrm{m/s^2})$ and enabled collision detection. A collision-enabled ground plane supported both the manipulator and target phantoms. The simulation models consist of both regular phantoms (rectangular prisms and cylinders) and irregular phantoms shows in Fig. 3.1 (b1,c1). The irregular phantoms are designed with either flat surfaces or small protrusions to mimic tumor masses. All phantoms are assigned collision properties to ensure realistic physical interactions.

Robotic system. The manipulator was modelled as a six-degree-of-freedom ED6 arm [19], imported from its URDF description at scale factor 1.0. Genesis enabled Jacobian computation and IK for this model. The base of the arm was fixed to the ground to emulate an industrial setup. Joint controllers were implemented in position mode with proportional–derivative (PD) gains and torque/velocity limits to maintain stability. To reflect realistic behaviour, the arm was returned to its home configuration after each detection or tracking trial using interpolated joint trajectories, rather than instantaneous resets.

Vision system. An RGB-D camera was modelled using the thin-lens configuration and rigidly mounted to the J6 link of the manipulator with a 10 mm offset along the z-axis to avoid self-occlusion. The camera provided 640 × 480 px RGB and depth images, with a field of view of $45°$, aperture $f/2.8$, and a focal length equivalent to 20 mm. Depth sensing operated reliably in the range 0.05–0.8 m, matching tabletop phantom scales. Bilateral filtering [26] was applied to reduce depth noise while preserving edges, and invalid or extreme depth values were discarded. This eye-in-hand setup emulated the principle of vision-based tactile sensors, where perception is coupled directly to the end-effector. Additional softaware support was provided by OpenCV [2], Open3D [32], and Scikit-learn [18].

2.3 Perception

Reliable palpation requires the probe to be placed accurately above the region of interest before any surface contact is attempted. This makes phantom localization the first and most critical step of the servoing pipeline. Without a precise estimate of the phantom centroid and spatial extent, the manipulator cannot be guided safely to initiate boundary tracking or subsequent tactile exploration.

2.3.1 HSV+Depth segmentation.

In this approach, the manipulator executed a 360° sweep around the workspace, capturing RGB-D frames from 24 evenly distributed viewpoints. Each RGB frame was converted into the HSV colour space [23], where binary masks were generated by thresholding the hue, saturation, and value channels. Thresholds were adaptively tuned to account for illumination conditions: under bright lighting the lower value bound was increased, whereas under dim conditions the saturation bound was relaxed to preserve detection sensitivity. Morphological opening with a 5×5 kernel [14] was applied to remove isolated noise pixels.

For each segmented region, the largest contour was selected, and its centroid pixel (u, v) was computed using image moments. Given a valid depth value d at this location and the camera intrinsics (f_x, f_y, c_x, c_y), the corresponding 3D camera-frame coordinates (x_c, y_c, z_c) were obtained by back-projection:

$$x_c = \frac{(u - c_x)\, d}{f_x}, \quad y_c = \frac{(v - c_y)\, d}{f_y}, \quad z_c = d. \tag{2.1}$$

These points were then transformed into world coordinates using the homogeneous camera-to-world extrinsic matrix T_{cw}:

$$\mathbf{p}_{world} = T_{cw} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}. \tag{2.2}$$

By aggregating detections across the 24 viewpoints, the method estimated the global centroid $\bar{\mathbf{p}}$ of the phantom:

$$\bar{\mathbf{p}} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{p}_{world}^{(i)}, \tag{2.3}$$

where N is the number of valid detections. The phantom's x/y extent was approximated by the minimum and maximum coordinates of all back-projected boundary points.

This approach is computationally lightweight and achieves near real-time performance. However, its reliance on colour cues makes it vulnerable to illumination variation and background clutter, motivating the introduction of depth-based clustering as a more robust alternative.

2.3.2 Depth+Clustering.

The second approach directly exploited the depth channel, thus avoiding colour sensitivity. Each depth frame was first filtered to retain points within a plausible workspace range (typically 0.15–0.60 m under normal lighting, with relaxed thresholds for bright and dim conditions). Ground-plane points were removed by applying a z-axis

cutoff of 2 mm:

$$\mathcal{P}_{valid} = \{(u, v, d) \mid d_{min} < d(u, v) < d_{max},\ z(u, v, d) > z_{th}\}\,. \tag{2.4}$$

Valid depth pixels were back-projected into 3D camera coordinates using the intrinsic matrix K:

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = d\,K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \qquad K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{2.5}$$

and subsequently transformed into the world frame by the extrinsic matrix T_{cw}.

The set of 3D points $\mathcal{P}_{world}$ was clustered using the density-based DBSCAN algorithm [7]. DBSCAN defines clusters as dense regions satisfying:

$$\text{dist}(\mathbf{p}_i, \mathbf{p}_j) \leq \varepsilon, \quad |\mathcal{N}_\varepsilon(\mathbf{p}_i)| \geq \text{minPts}, \tag{2.6}$$

where ε is the neighbourhood radius and minPts the minimum samples required to form a core point. Parameters were tuned adaptively for each lighting condition (e.g., $\varepsilon = 0.03$, minPts $= 5$ under normal lighting).

The largest non-noise cluster $\mathcal{C}^*$ was selected as the target phantom. Its centroid $\bar{\mathbf{p}}$ and bounding box were computed as:

$$\bar{\mathbf{p}} = \frac{1}{|\mathcal{C}^*|} \sum_{\mathbf{p} \in \mathcal{C}^*} \mathbf{p}, \qquad x_{min} = \min_{\mathbf{p} \in \mathcal{C}^*} x,\ x_{max} = \max_{\mathbf{p} \in \mathcal{C}^*} x,\ \text{etc.} \tag{2.7}$$

By aggregating detections across the 24 viewpoints, the method produced a stable estimate of the phantom's centroid and spatial extent. While this approach incurs higher computational cost compared with HSV+Depth segmentation, it provides more accurate and consistent localisation across varied lighting conditions and irregular tissue geometries.

2.4 Boundary Construction

After estimating the phantom's position and lateral extent, the next task was to delimit a safe region for surface exploration. Instead of probing directly across the estimated bounding box, which risks overshooting beyond the phantom, we reconstructed a closed boundary curve around the target. The safe region is then defined as the interior of this curve.

To achieve this, we implemented two boundary detectors—Canny, RGB-D fusion. Although they differ in how boundary candidates are extracted, all follow the same iterative loop: (i) boundary points are detected from RGB-D images; (ii) these points are projected into 3D and accumulated into a partial point cloud; (iii) the manipulator advances step-by-step along the detected boundary; and (iv) once the accumulated point cloud satisfies closure criteria, the safe polygon is accepted for interior scanning.

2.4.1 Boundary detection

Canny edges In the first method, each RGB image captured by the end-effector camera was converted to grayscale and processed with the Canny operator [20]. This operator applies a Gaussian smoothing followed by gradient computation and non-maximum suppression, before applying dual thresholds. The output is a binary edge map highlighting intensity discontinuities in the image.

Contours were then extracted using a chain-approximation algorithm [28]. Unlike traditional Canny-based

segmentation where small contours are discarded, all detected contours were retained to avoid losing legitimate boundary segments. Each contour pixel was paired with its corresponding depth value and back-projected into 3D using the camera intrinsics. These 3D points were accumulated across successive viewpoints to incrementally grow the boundary point cloud.

RGB-D fusion In the RGB-D method, boundaries were extracted directly from depth discontinuities. Depth maps were preprocessed to remove invalid ranges and denoised using **bilateral filtering** [26]. The filtered map was normalised to 8-bit, and **Sobel operators** [12] (kernel size 5) were applied to compute horizontal and vertical gradients, producing gradient maps that emphasised depth changes. To ensure geometric consistency,the **gradient magnitude** image was then thresholded to isolate strong discontinuities, followed by **morphological** opening and closing to remove small noise and reconnect fragmented edges. Finally, external contours were traced from the resulting binary map to obtain a set of boundary curves. These contours, once projected into world coordinates using camera intrinsics and extrinsics, were integrated into the iterative scanning loop that incrementally advanced the manipulator along the detected edge.

2.4.2 Boundary tracking and point cloud accumulation

Once boundary points were extracted by the RGB-D detector, the manipulator advanced iteratively along the object boundary using a ”jumping” strategy. At each step, the system selected the farthest detected boundary point relative to the current end-effector position as the next waypoint, subject to clockwise direction constraints to ensure systematic coverage. The selected waypoint was then projected to a predefined height above the object surface (0.15m base height with adaptive adjustment up to 0.25m). Before executing the movement, a fresh RGB-D frame was captured and boundary points were extracted from the current view. The arm then executed a translational ”jump” to the target waypoint using IK and path planning. This process was repeated until either the arm returned to the vicinity of its starting position (within 3cm threshold), closure criteria were met through point cloud analysis, or the maximum number of steps (40) was reached.

2.4.3 Closure Verification

To ensure that a complete loop had been captured, a hybrid closure check was applied to the accumulated boundary cloud. Two complementary tests were run in parallel:

The first test was image-based: the 3D boundary points were projected onto the xy plane and discretised into a 2D grid. After applying morphological thickening to close small gaps, a **flood-fill operation** was launched from outside the shape. Closure was declared if the remaining “inland” area exceeded a minimum fraction of the phantom’s estimated top surface, ensuring that the detected boundary enclosed a meaningful region.

The second test was graph-based: a k-**nearest neighbour graph** was constructed over the projected boundary points, and connectivity was analysed. A valid loop was confirmed if the graph contained cycles with most nodes having degree two, and if the maximum edge length remained below an adaptive threshold relative to point spacing.

In practice, multiple loops could appear, for example when local protrusions formed secondary contours. To avoid prematurely accepting such inner loops, we retained the **outermost boundary** (maximum projected area) as the valid contour, and required its area to exceed a fraction τ of the prior estimate A_{prior} obtained from Stage 1 perception. The final closure decision was made in a permissive manner: if either the image-based or the graph-based test succeeded on this outer loop, the boundary was accepted as closed. At this point, the polygon was

designated as the **safe region**, and scanning transitioned to the subsequent surface reconstruction stage.

2.5 Surface Scanning and DEM Reconstruction

Once the phantom boundary was successfully closed, a dense surface scan was conducted inside the safe region (closed polygon). The aim of this stage was to obtain a high-fidelity point cloud of the surface, reconstruct its height distribution, detect tumour-like protrusions, and convert them into prioritized palpation targets.

2.5.1 Surface Point-Cloud Acquisition

The polygonal boundary (safe region) obtained from the last step was projected onto the xy-plane. The interior was then rasterised into a uniform grid of candidate sampling locations. To guarantee that only valid points were considered, each grid position was filtered using a point-in-polygon test to confirm inclusion within the boundary polygon. The manipulator was then guided above each valid location using IK, maintaining a constant vertical offset (approximately 0.12 m) to avoid physical contact during scanning. At each pose, an RGB-D frame was captured and converted into a world-frame point cloud. Only surface points lying within a small depth window were retained to form a clean surface layer. Across all grid positions, the accumulated surface points were combined, down-sampled, and de-duplicated to balance density and computational load.

2.5.2 DEM Reconstruction

Although the accumulated surface point cloud provided a direct representation of the phantom's geometry, the points were distributed irregularly due to occlusions and varying camera viewpoints. Such uneven density made it difficult to perform local neighbourhood analysis reliably. To address this, these points were interpolated onto a regular two-dimensional lattice, forming a DEM of surface height $Z(x, y)$ across the region of interest. The grid resolution was chosen adaptively based on the spatial extent of the scanned region, while an upper bound on grid size was enforced to maintain computational efficiency. Interpolation was performed using scattered-data methods (Delaunay-based linear interpolation), which preserved local geometry without extrapolating into unsupported areas [24]. This produced a smooth and compact surface representation, on which subsequent neighbourhood-based operations (e.g. anomaly detection kernels) could be applied reliably.

2.6 Anomaly Detection and Palpation Suggestions

With the DEM reconstructed from the surface point cloud, the system proceeded to analyse local variations in surface height in order to identify protrusions resembling tumour surrogates. For each cell in the DEM, the surrounding neighbourhood was examined, and the local height range was computed. Regions where this variation substantially exceeded the global background distribution were flagged as anomalies. An adaptive threshold was employed so that the detection sensitivity adjusted automatically to the roughness of the surface under examination.

Detected anomalies were then refined through connected-component analysis, which grouped neighbouring above-threshold cells into coherent regions. Each region was characterised by its centroid, estimated height elevation, surface area, and relative strength, thereby providing a quantitative description of suspicious features on the surface. Because this approach relied solely on 3D geometry, it was unaffected by illumination or texture, and was therefore robust across different simulated conditions.

To translate these findings into actionable steps for the manipulator, the centroids of the most prominent anomaly regions were designated as probing waypoints, placed at a safe stand-off height to avoid premature contact. The anomalies were ranked by their relative strength, and only the top candidates were retained to limit the probing sequence. In cases where no anomaly was detected, a fallback waypoint at the phantom centre was generated to ensure palpation could still proceed. The resulting set of palpation suggestions thus provided the bridge between vision-based surface analysis and the subsequent tactile exploration stage.

Chapter 3

Result

3.1 Key Results

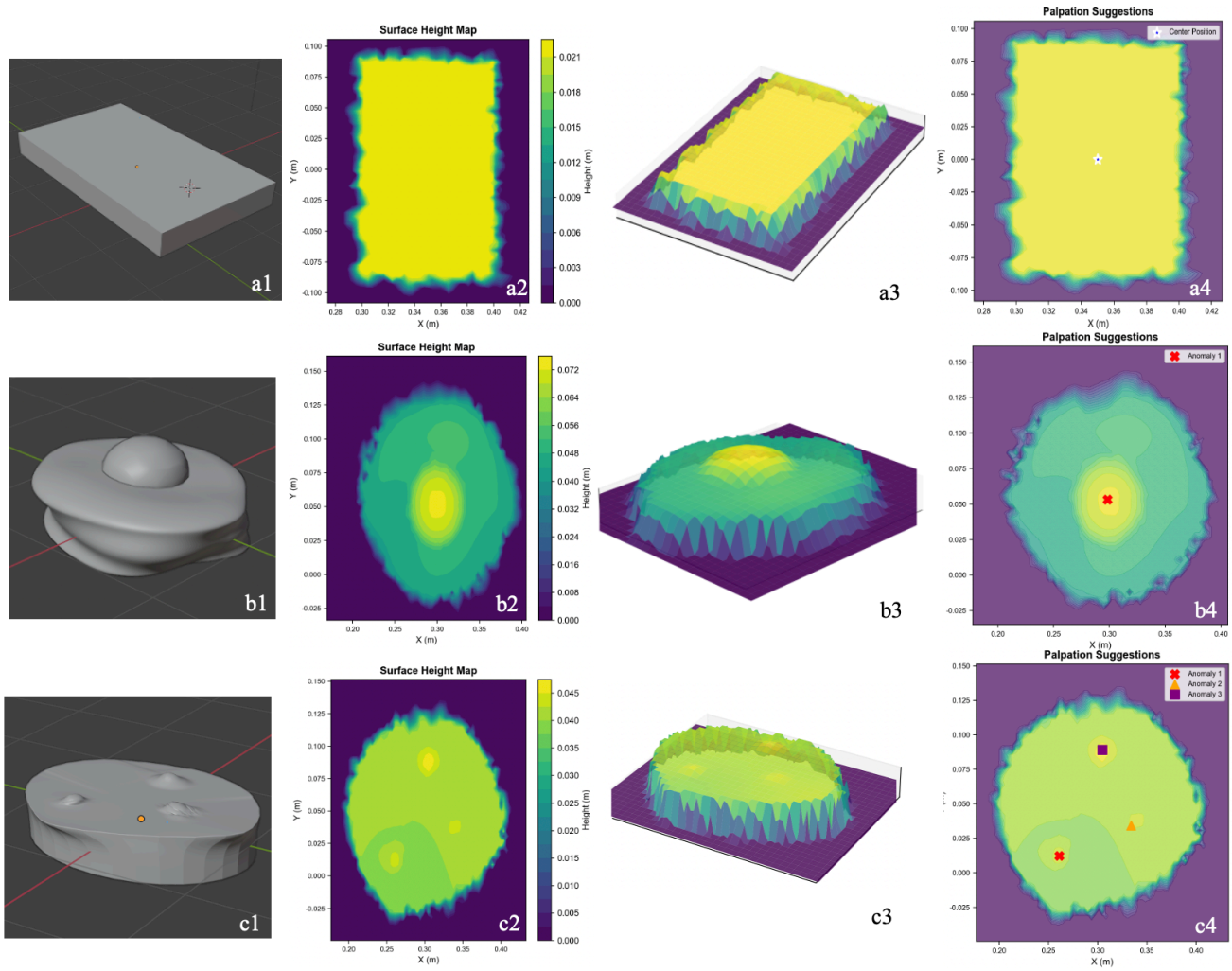

Figure 3.1: **Surface scanning and palpation suggestion results.** Top row (a1–a4): regular cuboid phantom; middle row (b1–b4): irregular phantom with protrusion; bottom row (c1–c4): irregular phantom with multiple protrusions. Columns show (1) phantom model, (2) reconstructed surface height map, (3) interpolated 3D surface model, and (4) palpation suggestions derived from detected anomalies.

For the final stage of qualitative pipeline evaluation, we tested both regular and irregular phantoms, present-

ing three representative categories: (a1) regular rectangular prism objects, (b1) irregular objects with a single protrusion simulating a tumour, and (c1) irregular objects with multiple localized protrusions.

For each phantom, reconstructed surface elevation maps are shown in Fig. 3.1(a2–c2), where local height variations can be clearly observed. These elevation maps were further interpolated into continuous three-dimensional surface models Fig. 3.1(a3–c3), forming DEMs suitable for anomaly analysis. In the final step, the system automatically detected tumour-like protrusions as anomalies and generated corresponding palpation suggestions Fig. 3.1(a4–c4). For objects with smooth, regular surfaces, the centre position was designated as the reference palpation site. For irregular phantoms, the algorithm successfully localised the designed protrusions and marked them as high-priority palpation targets. The system also outputs the corresponding palpation coordinates and priority levels at the terminal.

These results demonstrate that the proposed pipeline not only reconstructs surface geometry reliably but also translates it into actionable palpation guidance, thereby establishing the crucial link between vision-based surface scanning and subsequent tactile verification.

3.2 Perception: Phantom Localisation

To initialise the palpation pipeline, two complementary localisation strategies were evaluated: (i) **HSV+Depth segmentation**, which uses colour thresholding followed by depth back-projection; and (ii) **Depth+Clustering**, which applies depth thresholding and DBSCAN clustering to extract the dominant object cluster.

Both methods were systematically tested across three illumination conditions (*normal, bright, dim*), with each condition repeated five times. In each trial, only a single phantom was placed in the workspace, with its position randomly perturbed within a fixed radius. Two phantom geometries were considered: a regular box phantom and an irregular phantom with surface protrusions. This yielded a total of 60 independent trials ($3 \times 5 \times 2 \times 2$). Ground-truth centroid and extent were known from the simulator, enabling quantitative evaluation of detection time, centroid error, and lateral range error. A trial was considered successful if both position and range errors were below 2 cm.

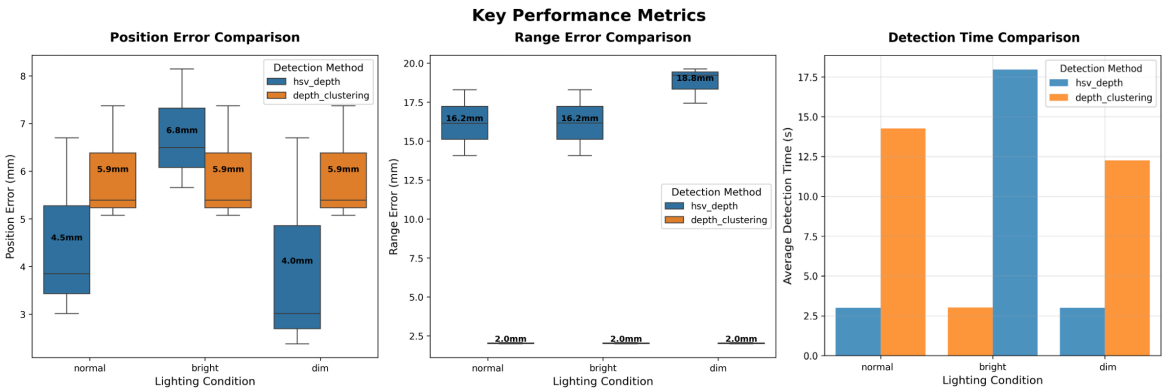

Figure 3.2: Comparison of localisation performance between HSV+Depth and Depth+Clustering across three illumination conditions. Results are averaged over five randomised trials per condition for both regular and irregular phantoms (total $n = 60$). Error bars indicate standard deviation.

Overall performance. Both approaches achieved a 100% success rate across all 60 trials, with centroid localisation errors consistently below 1 cm. However, their efficiency and range precision differed markedly (Fig. 3.2). HSV+Depth was lightweight, converging in an average of 3.0 ± 0.4 s across all conditions. Its centroid error averaged 5.7 ± 1.5 mm, but lateral extent estimation was less precise (19.5 ± 2.0 mm error).

Depth+Clustering required substantially longer computation (14.8 ± 2.4 s on average) but consistently achieved sub-centimetre centroid error (5.5 ± 0.3 mm) and highly accurate range estimation (2.1 ± 0.1 mm).

Impact of illumination. Under bright conditions, HSV+Depth produced elevated centroid errors ($\approx$ 7 mm) as the HSV thresholding became less reliable. By contrast, Depth+Clustering maintained uniform accuracy but required longer run-times in dim conditions due to relaxed depth thresholds and increased false clusters.

Summary. Both localisation strategies achieved sub-centimetre centroid accuracy and 100% success rates across all trials. However, their trade-offs were clear: HSV+Depth was consistently faster ($\approx 3, s$) but suffered from large range errors ($\approx 20, mm$), while Depth+Clustering required longer runtimes ($\approx 15, s$) yet produced highly reliable extent estimates ($\approx 2, mm$). Since accurate range estimation is critical for defining the safe scanning region in subsequent boundary tracking and surface palpation, the final pipeline adopted the **Depth+Clustering** method as the default localisation strategy, ensuring robustness across both lighting conditions and object geometries.

3.3 Boundary Tracking

Boundary reconstruction was assessed using two methods: **Canny edge detection** and **RGB-D gradient fusion**. To evaluate performance, we compared the reconstructed boundary clouds against simulator ground truth using a set of complementary metrics (Fig. 3.3), while detailed per-object and per-condition results are provided in Appendix A.

Specifically, **Chamfer distance** measures the bidirectional mean distance between detected boundary points and the ground-truth contour; smaller values indicate that the detected contour lies closer on average to the true shape. **Hausdorff distance** captures the maximum boundary deviation, reflecting worst-case errors; again, lower values are preferable. To assess stability, we further measured the **standard deviation of residuals** and the **coefficient of variation (CV)** across trials. These quantify the spread and relative variability of the boundary errors, with smaller values indicating more consistent reconstruction. In addition, **execution time** was recorded to quantify efficiency, and the **point count** served as a proxy for boundary density and completeness.

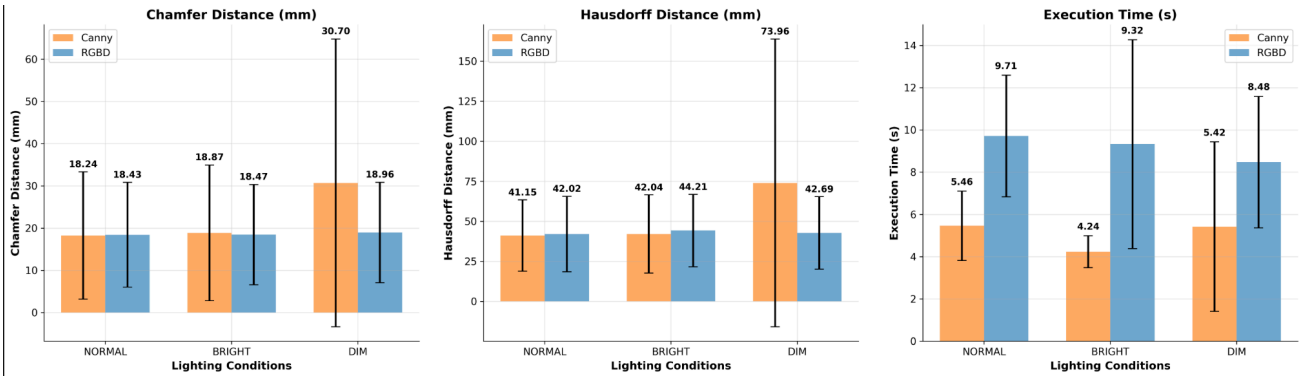

Figure 3.3: Comparison of boundary tracking performance between Canny and RGB-D fusion across three illumination conditions. Metrics shown are Chamfer distance, Hausdorff distance, and execution time (mean $\pm$ standard deviation).

Results. Across all trials, RGBD consistently outperformed Canny in contour accuracy. The mean Chamfer distance improved from 22.6 mm (Canny) to 18.6 mm (RGBD, 17.6% improvement), and the Hausdorff distance reduced from 52.4 mm to 43.0 mm (18.0% improvement). Stability was also better: residual spread decreased from 12.8 mm to 9.6 mm, and CV fell by 27.8%. RGBD produced denser and more continuous boundaries (over

9,000 points vs. ≈4,500 for Canny), improving surface coverage (0.851 vs. 0.823).

The trade-off was computation time: Canny was faster (5.0 s vs. 9.2 s), making it suitable for lightweight applications but prone to fragmentation in challenging cases (e.g. irregular phantoms under dim lighting, where Hausdorff >130 mm). RGBD remained stable across all conditions, including irregular phantoms with protrusions.

Summary. Overall, RGBD provided *denser, more accurate, and more robust boundary reconstructions*, while Canny was faster but less reliable. Given that boundary completeness is critical for subsequent surface scanning and anomaly detection, the pipeline ultimately adopted **RGBD boundary tracking** as the default method.

Chapter 4

Conclusion

This work develops a vision-guided *pre-palpation support* pipeline that bridges phantom perception and tactile verification in a unified simulation framework.

A five-stage framework was implemented and validated in the Genesis simulator, integrating phantom localisation, boundary reconstruction, closure verification, surface scanning, and anomaly detection into a coherent workflow. For perception, two complementary strategies were compared: HSV+Depth segmentation offered fast performance but proved sensitive to illumination, whereas Depth+Clustering provided more robust and accurate estimates, ultimately serving as the default method. For boundary construction, RGB-D fusion outperformed Canny edge detection, producing denser and more continuous contours that defined safer scanning regions. Closure verification combined image-plane and graph-based tests to guarantee valid loops, while surface scanning converted point clouds into DEMs, enabling local protrusions to be detected as tumour surrogates. These were then translated into actionable palpation suggestions.

Overall, the pipeline successfully bridged vision-based perception with tactile exploration. By providing both a safe region for surface scanning and concrete palpation suggestions, it ensured that subsequent probing could be performed more efficiently, reliably, and safely.

Limitations and Future Work Although the proposed pipeline demonstrated stable performance in simulation, several limitations remain. The simulated phantoms were relatively simple, consisting mainly of cuboids and cylinders with protrusions to mimic tumours, which does not fully capture the heterogeneity of real soft tissues. Similarly, the RGB-D camera model and lighting variations were restricted, leaving a gap between simulated conditions and those encountered in real experiments or surgical environments. Most importantly, the system and pipeline have not yet been deployed on a physical robotic platform for end-to-end validation.

Future work will therefore focus on extending the framework towards greater realism and practical applicability. More complex and irregular phantoms with diverse tumour morphologies will be introduced, and integration with physical vision–tactile sensors such as TacTip or ViTacTip will be pursued to evaluate simulation-to-reality transfer. In parallel, deep learning methods will be explored for end-to-end anomaly detection and palpation planning, while computational efficiency will be further optimised to support real-time robotic applications.

Chapter A

Appendix

A.1 Full Boundary Tracking Results

To complement the summary metrics in Section 3.3, the full condition-wise results for both cube and irregular phantoms are provided here. These include Chamfer distance, Hausdorff distance, residual spread, coefficient of variation, execution time, and point cloud density across all lighting conditions.

COMPARISON TABLE (Mean Values)

Object	Lighting	Method	chamfer_	hausdorf	std_resi	cv_resid	executio	point_co
Cube	normal	CANNY	3.179	18.966	4.273	1.612	3.935	3707.333
Cube	normal	RGBD	6.071	18.794	4.459	0.974	7.106	7474.667
Cube	bright	CANNY	2.827	17.984	3.868	1.623	4.218	3777.000
Cube	bright	RGBD	6.690	21.710	5.083	1.074	7.147	7132.667
Cube	dim	CANNY	3.117	17.736	4.279	1.543	4.187	3811.667
Cube	dim	RGBD	7.176	20.133	4.495	0.941	5.727	6015.000
ctry.obj	normal	CANNY	33.297	63.341	14.311	0.666	6.992	6117.333
ctry.obj	normal	RGBD	30.787	65.253	14.593	0.643	12.318	11478.000
ctry.obj	bright	CANNY	34.909	66.088	14.705	0.620	4.256	3544.000
ctry.obj	bright	RGBD	30.250	66.719	14.616	0.641	11.502	11600.333
ctry.obj	dim	CANNY	58.273	130.186	35.480	0.714	6.654	5743.000
ctry.obj	dim	RGBD	30.739	65.248	14.054	0.619	11.232	10673.333

Figure A.1: Full comparison of boundary tracking results across object type, lighting condition, and method (Canny vs. RGBD).

A.2 Source Code Availability

All source code developed for this project is available at the following repository:

- https://github.com/JetfiremaxYao/MFP

The repository contains implementations of all major modules, including:

- Phantom localisation (HSV+Depth, Depth+Clustering)
- Boundary construction (Canny, RGB-D fusion, Alpha-Shape)

- Closure verification
- Surface scanning and DEM reconstruction
- Anomaly detection and palpation suggestion
- Unified evaluation framework

This repository ensures full reproducibility of the results presented in this report.

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